Module 3: Text Functions

3.1 Concept

Text functions in DAX are used to **clean, format, and manipulate string values**.
In our dataset (`AdvancedDAX_SalesData.csv`), text columns like **Product** and **Category** provide opportunities to:

- Create **labels** for visuals
- Build **codes/IDs**
- Standardize casing
- Search and replace text

3.2 Key Functions Overview

Function	What it Does	Example (on dataset)
`CONCATENATE(t1, t2)`	Joins text strings	`CONCATENATE(Sales[Category], " - " & Sales[Product])`
`LEFT(text, n)`	Extracts first *n* characters	`LEFT(Sales[Product], 3) → "Lap"`
`RIGHT(text, n)`	Extracts last *n* characters	`RIGHT(Sales[Product], 2) → "es"` for Shoes
`MID(text, start, num_chars)`	Extracts middle portion	`MID(Sales[Product], 2, 4)`
`LEN(text)`	Returns number of characters	`LEN(Sales[Product])`

Function	What it Does	Example (on dataset)
`TRIM(text)`	Removes leading/trailing spaces	`TRIM("  Bag ") → "Bag"`
`UPPER(text)`	Converts to uppercase	`UPPER(Sales[Category])`
`LOWER(text)`	Converts to lowercase	`LOWER(Sales[Category])`
`SEARCH(find, text)`	Finds substring (case-insensitive)	`SEARCH("Phone", Sales[Product])`
`FIND(find, text)`	Finds substring (case-sensitive)	`FIND("Phone", Sales[Product])`
`SUBSTITUTE(text, old, new)`	Replaces part of text	`SUBSTITUTE(Sales[Product], "Shoes", "Footwear")`

3.3 Practical Examples

Example 1: Full Product Label

```
Product Full Name :=
CONCATENATE(Sales[Category], " - " & Sales[Product])
```

- Creates `"Electronics - Laptop"`.

Example 2: Short Product Code

```
Product Code :=
LEFT(Sales[Product], 3) & "-" & RIGHT(Sales[Category], 2)
```

- `"Lap-ns"` for Laptop in Electronics.

Example 3: Detect Phones

```
Is Phone :=
IF(
    SEARCH("Phone", Sales[Product],,0) > 0,
    "Yes",
    "No"
)
```

- Returns `"Yes"` for Phone/Headphones, `"No"` for others.

Example 4: Standardized Category

```
Category Upper :=
UPPER(Sales[Category])
```

- Ensures consistent formatting for categories.

Example 5: Clean Product Names

```
Product Clean :=
SUBSTITUTE(Sales[Product], "Shoes", "Footwear")
```

🔹 Replaces `"Shoes"` with `"Footwear"`.

3.4 Exercises (Hands-On)

1. **Order Label**
   Create a column: `"Order-<SalesID>"`
   Example: `Order-15`.
2. **Category Initials**
   Use `LEFT + UPPER → "Electronics" = "E"`.
3. **Product Length**
   Count how many characters are in each product name.
4. **Find Fashion Products**
   Flag rows where `Category` contains `"Fashion"`.
5. **Trim Category**
   Add artificial spaces to categories and clean them with `TRIM`.

3.5 Visualizations in Power BI

1. **Table**

- Product | Product Full Name | Product Code | Category Initials

2. **Column Chart**
   - X-axis = Is Phone (Yes/No)
   - Y-axis = Total Sales
3. **Card Visual**
   - Show the longest product name (using `MAXX(LEN(Product))`).

3.6 Exercise Answers (For Trainers)

1. **Order Label**

```
Order Label :=
"Order-" & Sales[SalesID]
```

- Produces values like `"Order-1"` , `"Order-200"` .

2. **Category Initials**

```
Category Initial :=
LEFT(UPPER(Sales[Category]), 1)
```

- `"Electronics" → "E"` , `"Fashion" → "F"` , `"Accessories" → "A"` .

3. **Product Length**

```
Product Length :=
LEN(Sales[Product])
```

- `"Laptop"` → 6, `"Headphones"` → 10.

4. **Find Fashion Products**

```
Is Fashion :=
IF(SEARCH("Fashion", Sales[Category],,0) > 0, "Yes", "No")
```

- Returns `"Yes"` for all Fashion items.

5. **Trim Category**

```
Category Clean :=
TRIM(Sales[Category])
```

- `"  Fashion "` → `"Fashion"`.

✅ By the end of Module 3, students will:

- Use **text manipulation functions** (`LEFT` , `RIGHT` , `MID` , `LEN`).
- Build **labels, flags, and codes** for reporting.
- Standardize casing and clean category/product names.